

TrES-1b

R-1147

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_220311 · created 2026-07-31T00:47:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459649.9415 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.022
Transit depth (Rp/Rs)^2	2.03 +/- 0.64 [%]
Orbital Inclination (inc)	85.8 +/- 1.13
Airmass coefficient 1 (a1)	0.981 +/- 0.014
Airmass coefficient 2 (a2)	0.0106 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#2 - [495, 268]
Optimal Aperture	5.15
Optimal Annulus	9.55
Transit Duration (day)	0.069 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.022 vs 0.1358 (deviation 0.32 σ)
Duration fit vs published	0.069 d vs 0.1042 d (deviation 1.74 σ)
Mid-time O-C	-6.6 min
Depth SNR	6.4
Residual scatter	1.42 %

Light curve

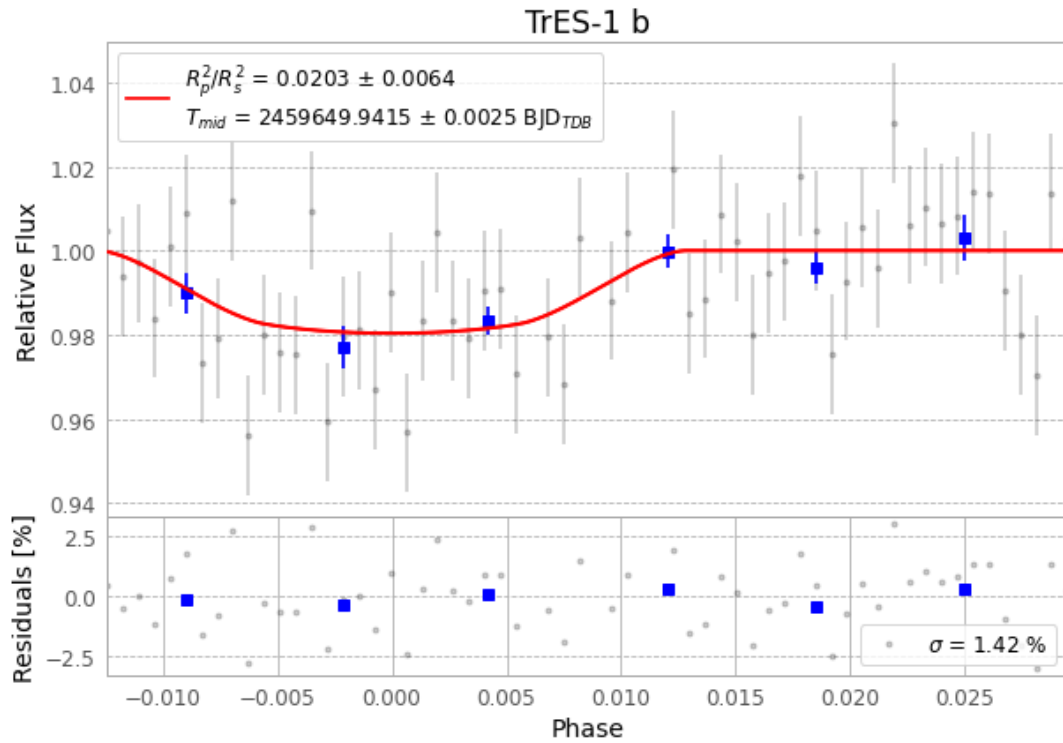


Figure 1 — FinalLightCurve_TrES-1 b_2022-03-11.png (SHA-256 066da67ec255959c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [203, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bcf6cfaab5936b3e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), TRES-1220311094212.FITS → TRES-1220311124512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-220311.log (7abb4ada30e2...), FinalLightCurve_TrES-1 b_2022-03-11.png (066da67ec255...), FinalLightCurve_TrES-1 b_2022-03-11.csv (04afaf7093b6...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:47Z.